Data Visualization & Analytics

CAPSTONE 2 — FINAL PROJECT REPORT

Global Retail Analytics: Uncovering Revenue Drivers, Customer Behavior, and Operational Patterns

Sector	Global E-Commerce & Retail
Team ID	SectionD — Team 10
Institute	Newton School of Technology
GitHub Repository	https://github.com/RISHIK92/SectionD_Team10_RetailAnalysis
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1. EXECUTIVE SUMMARY

This report presents the findings of a comprehensive data analytics project conducted on a global retail transactions dataset spanning 2023–2024. The project was carried out as part of the Data Visualization & Analytics (DVA) Capstone 2 at Newton School of Technology by Section D, Team 10.

BUSINESS PROBLEM

Global retail businesses generate vast volumes of transactional data but often lack the analytical infrastructure to translate raw records into actionable intelligence. Without structured analysis, retailers are unable to identify which product categories or geographies drive the most revenue, whether customer segmentation strategies are effective, or whether operational variables such as shipping methods and payment modes materially influence satisfaction or spend. This project was designed to close that gap.

APPROACH

The team sourced a 302,010-row retail dataset from Kaggle, covering five countries (USA, UK, Germany, Australia, Canada), five product categories, and 30 attribute columns. A four-stage Python pipeline — extraction, cleaning, EDA, and statistical analysis — was executed in Jupyter notebooks, followed by Tableau dashboard development for interactive visualization. The pipeline reduced the raw dataset to 293,468 clean, validated records across 33 engineered columns.

KEY FINDINGS

- Electronics leads all product categories with $95.19M in total revenue and the highest average transaction value ($1,370.83), yet the spread across all five categories is only ~$5 — indicating a uniformly priced product portfolio rather than category-driven revenue stratification.
- The USA contributes 31.6% of total revenue ($126.92M), but per-transaction spend is virtually identical across all five markets (median ~$1,042), demonstrating that the USA's revenue dominance is driven by transaction volume, not premium pricing.
- Customer satisfaction is stable at 49.1% (Ratings >= 4) across the full dataset, with Premium segment customers showing the highest satisfaction rate at 56.2% — suggesting a quality-of-experience advantage that justifies loyalty investment.
- Monthly revenue shows minimal seasonality (~6% swing between peak and trough months), enabling stable year-round inventory and workforce planning without the need for seasonal promotional peaks.

KEY RECOMMENDATIONS

- Prioritize customer acquisition in Australia and Canada — both markets already match USA per-transaction values ($1,372 and $1,366 respectively) but account for only 15% and 14.9% of revenue, representing the highest-yield growth opportunity through volume expansion.
- Invest in Electronics category depth and cross-category bundling with Grocery (the second-highest volume category), as Electronics generates the highest revenue despite having the fewest SKUs — indicating strong demand elasticity.
- Launch a Premium segment onboarding programme: Premium customers are the most satisfied (56.2%) but contribute only 21.1% of revenue. Accelerating the path from 'New' to 'Premium' through targeted loyalty milestones could shift the revenue mix significantly.

2. SECTOR & BUSINESS CONTEXT

SECTOR OVERVIEW

The global e-commerce and retail sector is one of the fastest-growing industries in the world. Multi-channel retail — spanning physical stores, online platforms, and hybrid fulfilment models — is increasingly data-intensive. Businesses must manage product catalogues across thousands of SKUs, serve customers across different time zones and income brackets, and route orders through multiple shipping and payment channels.

The five markets represented in this dataset — USA, UK, Germany, Australia, and Canada — collectively represent over $2 trillion in annual retail spending. Retailers operating across these markets face challenges including demand forecasting, inventory optimization, customer retention, and satisfaction monitoring.

DECISION-MAKER PROFILE

The primary audience for this project is the Chief Commercial Officer (CCO) or Head of Analytics at a mid-to-large multi-national retail firm. Secondary audiences include category managers, regional sales leads, and customer success teams who require actionable, segment-level intelligence.

BUSINESS VALUE

By converting raw transaction logs into structured KPIs and interactive dashboards, this project enables retail leaders to: (1) monitor revenue trends in real time; (2) compare performance across geographies and product categories without manual data wrangling; (3) assess the impact of customer segmentation on revenue and satisfaction; and (4) make operationally grounded decisions about shipping, payment, and inventory strategy.

3. PROBLEM STATEMENT & OBJECTIVES

FORMAL PROBLEM DEFINITION

Given a global retail transactions dataset of 302,010 records across 30 columns, can we identify the primary drivers of revenue, customer satisfaction, and operational efficiency — and translate those findings into a decision-ready Tableau dashboard with statistically validated insights?

PROJECT OBJECTIVES

- Clean and validate a raw, messy retail dataset using a reproducible Python ETL pipeline.
- Compute and define a KPI framework (Total Revenue, AOV, Satisfaction Rate, High-Value Flag) that maps to real business decisions.
- Conduct exploratory and statistical analysis to identify which features (geography, demographics, category, shipping, payment) have a material impact on revenue and satisfaction.
- Build an interactive Tableau dashboard (3 views) that allows non-technical stakeholders to explore the data by country, category, segment, and time period.
- Produce 8–12 actionable insights and 3–5 prioritized recommendations with estimated business impact.

SCOPE

In scope: Transaction-level analysis, customer segmentation, geographic revenue breakdown, product category performance, shipping and payment method analysis, and time-series revenue trends for 2023–2024.

Out of scope: Real-time data ingestion, predictive modelling, social media sentiment, supply chain cost data, and individual customer PII (removed during cleaning).

SUCCESS CRITERIA

- A clean dataset with zero null values and zero duplicates, confirmed by validation notebooks.
- Statistical tests performed on all major variable pairs with clear interpretation of practical vs. statistical significance.
- Tableau dashboards accessible on Tableau Public with at least 3 interactive views.
- Report delivering 8+ data-backed insights and 3+ quantified recommendations.

4. DATA DESCRIPTION

DATASET SOURCE

Dataset: Global E-Commerce Retail Transactions Dataset

Source: Kaggle (publicly available synthetic retail transactions dataset)

GitHub Path: data/raw/raw_retail_data.csv

The dataset simulates transactional records for a multi-national retail business operating across five countries, five product categories, and multiple customer segments. It was selected for its breadth of demographic, geographic, and operational attributes — making it suitable for a full-spectrum retail analytics project.

DATA STRUCTURE

Attribute	Value
Raw Rows	302,010
Raw Columns	30
Time Period	2023–2024 (Partial)
Countries	USA, UK, Germany, Australia, Canada
Product Categories	Electronics, Grocery, Clothing, Books, Home Decor
Customer Segments	New, Regular, Premium
Clean Rows (post-ETL)	293,468
Final Columns (post-ETL)	33 (25 original + 8 engineered)

KEY COLUMN DESCRIPTIONS

Column	Type	Description
Transaction_ID	Integer	Unique identifier for each retail transaction
Customer_ID	Integer	Unique customer identifier (anonymised)
Country / City / State	Categorical	Geographic location of the transaction
Age	Integer	Customer age at time of transaction

Gender	Categorical	Customer gender (Male / Female)
Income	Categorical	Income bracket: Low, Medium, High
Customer_Segment	Categorical	Loyalty tier: New, Regular, Premium
Date / Year / Month	Date/Time	Transaction date with derived time components
Total_Purchases	Integer	Number of items in the transaction (1–10)
Amount	Float	Per-item price for the transaction
Total_Amount	Float	Total transaction value (Amount x Total_Purchases)
Product_Category	Categorical	High-level category (5 categories)
Product_Brand	Categorical	Brand name associated with the product
Product_Type	Categorical	Sub-category product type
Feedback	Categorical	Customer feedback label: Excellent, Good, Average, Bad
Ratings	Integer	Numeric rating 1–5 (maps directly to Feedback)
Shipping_Method	Categorical	Express, Same-Day, or Standard
Payment_Method	Categorical	Cash, Credit Card, Debit Card, PayPal
Order_Status	Categorical	Delivered, Shipped, Processing, Pending

DATA LIMITATIONS & KNOWN BIASES

- The dataset is synthetic — it simulates retail transactions rather than representing actual business data. Patterns found may not generalise to real-world retail operations.
- Gender is binary (Male/Female only), which may not reflect the full diversity of customer demographics.
- 2024 data is partial (January–February only), making direct year-on-year comparisons unreliable without normalisation.
- Income brackets are self-reported or imputed labels and may not correlate with actual disposable income in each market.
- Feedback and Ratings are near-deterministically linked (Bad = Rating 1, Excellent = Rating 4-5), limiting independent analytical value of each variable.

5. DATA CLEANING & ETL PIPELINE

All cleaning and transformation steps were executed in Python using pandas. The pipeline is fully documented in notebooks/02_cleaning.ipynb, committed to the GitHub repository.

STEP 1 — COLUMN REMOVAL

Five personally identifiable columns were removed: Name, Email, Phone, Address, and Zipcode. These fields carry no analytical value for the objectives of this project and represent a privacy risk. The dataset was reduced from 30 to 25 columns.

STEP 2 — DUPLICATE REMOVAL

Two rounds of deduplication were performed. First, 4 fully identical rows were dropped using drop_duplicates(keep='first'). Second, 7,544 rows with duplicate Transaction_IDs (different row data, same ID — indicating data generation errors) were removed using subset=['Transaction_ID']. In total, 7,548 rows were removed, leaving 294,462 records.

STEP 3 — DATA TYPE CONVERSIONS

- Transaction_ID and Customer_ID were cast from float64 to Int64 (nullable integer) to correctly handle NaN values.
- Date was parsed with pd.to_datetime(), enabling extraction of Year, Month, and time-based features.
- Age, Year, Total_Purchases, and Ratings were cast to Int64 for correct numerical handling.
- Amount and Total_Amount were rounded to 2 decimal places.
- 14 categorical columns (City, State, Country, Gender, Income, Customer_Segment, Product_Category, Product_Brand, Product_Type, Feedback, Shipping_Method, Payment_Method, Order_Status, products) were cast to Pandas category dtype for memory efficiency.
- Month was set as an ordered categorical variable (January through December) to ensure correct sorting in downstream analysis.

STEP 4 — MISSING VALUE TREATMENT

Missing value counts before treatment ranged from 169 (Age) to 354 (Total_Purchases). The following strategy was applied:

- Rows with missing Transaction_ID, Customer_ID, or Amount were dropped (critical key/value fields).
- Numerical columns (Age, Ratings): filled with column median.
- Categorical columns (Gender, Customer_Segment, Product_Category, Product_Brand, Feedback, Shipping_Method, Payment_Method, City, State, Country, Income): filled with column mode.
- Total_Purchases / Amount / Total_Amount (695 rows with partial data): a consistency-based imputation was applied — missing values were derived algebraically where two of the three fields were available (e.g., Total_Amount / Amount = Total_Purchases). Remaining gaps were filled with the column median, and Total_Amount was recalculated as Amount x Total_Purchases for consistency.
- Date-related nulls (351 rows): rows with missing Date were dropped, and Year/Month were re-derived from the parsed Date column. Missing Time values were filled with '00:00:00'.

STEP 5 — OUTLIER DETECTION & TREATMENT

Outlier detection using the IQR method (1.5x rule) was applied to Amount, Total_Amount, and Total_Purchases. Results: Amount (0 outliers), Total_Purchases (0 outliers), Total_Amount (3,777 outliers = 1.29%). Given the dataset size (293K rows) and the nature of the variable (total transaction value), these high-value transactions were assessed as legitimate premium purchases rather than data errors. No rows were removed for outliers.

STEP 6 — FEATURE ENGINEERING (ETL LOAD PREP)

Eight new columns were added in notebook 05_final_load_prep.ipynb to support Tableau visualisation and KPI computation:

New Column	Definition
Month_Num	Numeric month (1=January, 12=December) for correct Tableau sort order
Quarter	Quarter label (Q1–Q4) derived from Date

Day_of_Week	Day name (Monday–Sunday) derived from Date
Hour	Hour of transaction (0–23) derived from Time column
Revenue_per_Purchase	Total_Amount / Total_Purchases — average basket value per item
High_Value_Flag	1 if Total_Amount > 75th percentile ($2,031.33), else 0
Satisfied_Flag	1 if Ratings >= 4, else 0 — binary satisfaction indicator
Feedback_Score	Numerical encoding: Excellent=3, Good=2, Average=1, Bad=0

FINAL DATASET STATE

After the complete ETL pipeline: 293,468 rows, 33 columns, zero null values, zero duplicate rows. All data types are correctly assigned. The file was exported as tableau_ready_dataset.csv for Tableau ingestion.

6. KPI & METRIC FRAMEWORK

The following KPIs were defined to structure the analytical narrative and drive the Tableau dashboard design. Each KPI maps at least one project objective and is computable from the cleaned dataset.

KPI	Formula	Business Purpose	Dashboard Value
Total Revenue	SUM(Total_Amount)	Absolute revenue benchmark across all filters	$401.6M
Avg Order Value (AOV)	MEAN(Total_Amount)	Measures basket size efficiency	$1,369
Revenue per Customer	Total Revenue / DISTINCT(Customer_ID)	Lifetime value proxy	$4,641.45
Total Orders	COUNT(Transaction_ID)	Transaction volume	293,468
Avg Units per Order	MEAN(Total_Purchases)	Cross-sell depth indicator	5.36
Satisfaction Rate	SUM(Satisfied_Flag) / COUNT(*)	% of orders rated 4 or 5	49.1%
Positive Feedback %	% Feedback = Excellent or Good	Sentiment benchmark	64.98%
High-Value Transaction %	SUM(High_Value_Flag) / COUNT(*)	Premium purchase rate	25.0%
Delivered Order Rate	% Orders with Status = Delivered	Fulfilment efficiency	43.52%
Revenue Share %	Country or Category Revenue / Total	Proportional contribution	By filter
High Income Revenue %	Revenue from Income=High / Total	Premium segment wallet share	24.8%
New vs Returning Split	New / (Regular + Premium) ratio	Customer base health	65%

7. EXPLORATORY DATA ANALYSIS

EDA was conducted in notebooks/03_eda_analysis.ipynb. All charts were generated using matplotlib and seaborn on the cleaned 293,468-row dataset.

7.1 UNIVARIATE ANALYSIS

Age Distribution

Age is broadly uniform between 18 and 70 years, with a mild right skew (skewness: 0.666) and negative kurtosis (-0.787), confirming a flat distribution with no dominant age cluster. The mean age is 35.3 years.

Amount Distribution

The Amount (per-item price) distribution is nearly perfectly symmetric (skewness: -0.003), ranging from $10 to $500 with a mean of $255.25. The platykurtic distribution (kurtosis: -1.199) indicates there is no dominant price point — the product portfolio is uniformly priced across its full range.

Total Amount Distribution

Total_Amount is right-skewed (skewness: 0.972), confirming most transactions are low-to-mid value ($439–$2,031 interquartile range) with a tail of high-value premium transactions. The 75th percentile threshold of $2,031.33 was used to define the High_Value_Flag KPI.

Ratings Distribution

Ratings spread evenly across 1–5 (negative kurtosis: -1.166), with a mean of 3.16. Each rating maps near-deterministically to a Feedback label (Bad=1, Average=2, Good=3, Excellent=4-5), confirming these two columns are informationally redundant.

7.2 CATEGORICAL DISTRIBUTION ANALYSIS

Gender

Male customers account for 62.2% of transactions, Female customers 37.8%. This male skew is consistent across all customer segments and countries, suggesting either a dataset artefact or a genuinely male-skewed customer base for the represented product mix.

Product Category

Electronics leads in transaction count (69,443 transactions), followed by Grocery (64,984), Clothing (53,212), Books (53,047), and Home Decor (52,782). The distribution is relatively even across non-Electronics categories, with Electronics maintaining a meaningful lead.

Payment Method

Credit Card is the most frequently used payment method (87,444 transactions, 29.8%), followed by Debit Card (75,181, 25.6%), Cash (70,989, 24.2%), and PayPal (59,854, 20.4%). Credit Card dominance aligns with typical e-commerce behaviour in the represented markets.

Order Status

Delivered accounts for 43.5% of all orders, shipped 21.5%, Processing 18.9%, and Pending 16.1%. The relatively high Processing and Pending proportions suggest significant pipeline inventory — potential for operational improvement.

Shipping Method

Same-Day is the most used method (101,535 transactions, 34.6%), marginally ahead of Express (99,441, 33.9%) and Standard (92,492, 31.5%). The near-uniform distribution indicates that customers are evenly distributed across shipping preferences.

7.3 BIVARIATE ANALYSIS

Product Category vs. Average Revenue

Electronics leads with an average transaction value of $1,370.83, followed by Clothing ($1,369.29), Home Decor ($1,368.48), Books ($1,367.60), and Grocery ($1,366.25). The spread across all categories is only ~$4.58, confirming uniform pricing across the product mix — no single category commands a pricing premium.

Age Group vs. Average Spend

The 36–45 age group shows the highest average spend ($1,404), while 18–25 shows the lowest ($1,347). However, the total spread is only ~$57 across all age groups, making age a statistically observable but practically negligible predictor of spend.

Income Level vs. Spend Distribution

All three income groups (Low, Medium, High) show nearly identical average spending (~$1,370). This counter-intuitive finding suggests either uniform product pricing across the income spectrum or a dataset limitation where income labels do not reflect actual purchasing power differences.

Top 10 Brands by Revenue

Pepsi dominates total brand revenue ($40.7M), nearly double the next brand, driven by high-volume Grocery transactions. Samsung ($24.7M) and Sony ($24.4M) lead in Electronics. Nike and Zara represent the Clothing/Sports segment. The non-grocery brand landscape is balanced, with no single brand accounting for disproportionate revenue.

Geographic Revenue

USA contributes the highest total revenue ($126.92M, 31.6%), followed by UK ($84.59M, 21.1%), Germany ($70.20M, 17.5%), Australia ($60.13M, 15.0%), and Canada ($59.80M, 14.9%). While the USA leads significantly in total revenue, the per-transaction median is essentially identical across all five markets (~$1,042), confirming that the USA's lead is volume-driven.

7.4 TIME-BASED ANALYSIS

Monthly Revenue Trend

Total monthly revenue remains remarkably stable throughout the year, ranging from $32.1M (February) to $34.3M (March/August) — a peak-to-trough swing of only ~6%. This absence of seasonality is a defining feature of this dataset and has significant implications for planning.

Year-over-Year

2023 accounts for approximately 84% of all transactions (~245,000 vs ~48,000 in 2024). The 2024 data cover only January and February, making direct year-on-year comparison unreliable without per-month normalization.

7.5 CUSTOMER SEGMENT ANALYSIS

Regular customers form the largest segment (48.7% of transactions), followed by New (30.1%) and Premium (21.1%). Counterintuitively, Premium customers have the lowest median transaction value ($1,040) compared to Regular ($1,044) and New ($1,042). This suggests segment classification reflects account tenure or loyalty tier rather than actual per-transaction spending power.

8. STATISTICAL ANALYSIS

All hypothesis tests were executed in notebooks/04_statistical_analysis.ipynb using scipy.stats. A significance level of alpha = 0.05 was used throughout. Given the large sample size (n = 293,468), most tests

achieve statistical significance even for negligible practical effects. For this reason, effect size measures (Cohen's d, Cramer's V, Spearman rho) are reported alongside p-values for correct interpretation.

8.1 PEARSON CORRELATION — AMOUNT AND TOTAL_PURCHASES VS. TOTAL_AMOUNT

Both Amount ($r = 0.67$, $p < 0.001$) and Total_Purchases ($r = 0.65$, $p < 0.001$) are strongly correlated with Total_Amount. This is expected as Total_Amount = Amount x Total_Purchases is a deterministic arithmetic relationship. Business implication: Any revenue forecasting model should use Amount and Total_Purchases as primary inputs.

8.2 SPEARMAN CORRELATION — AGE VS. RATINGS

Spearman rho = 0.17, $p < 0.05$. H0 rejected. Older customers give marginally higher ratings, but the effect size is negligible and not actionable. Business implication: Age-based satisfaction targeting is not supported by this data.

8.3 GENDER VS. TOTAL_AMOUNT — WELCH'S T-TEST

Male mean: $1,363 | Female mean: $1,373. t-statistic: statistically significant at n = 293K. Cohen's d < 0.01 (negligible). H0 rejected statistically, but the practical difference is less than $10 per transaction. Business implication: Gender-based pricing or promotional strategies targeting spend uplift are not supported.

8.4 INCOME LEVEL VS. TOTAL_AMOUNT — KRUSKAL-WALLIS TEST

Low mean: $1,372 | Medium mean: $1,370 | High mean: $1,361. H = statistically significant. Practical spread: ~$11 across income groups. Business implication: Income-tiered pricing strategies would yield no measurable revenue uplift.

8.5 CUSTOMER SEGMENT VS. TOTAL_AMOUNT — KRUSKAL-WALLIS TEST

Regular median: $1,044 | New median: $1,042 | Premium median: $1,040. Practical spread: $4. H0 rejected statistically due to sample size. Business implication: Segment labels reflect loyalty/tenure, not spending power. Cross-sell strategies should not rely on segment labels for spend targeting.

8.6 PRODUCT CATEGORY VS. TOTAL_AMOUNT — KRUSKAL-WALLIS TEST

Electronics leads with mean $1,370.83 but the inter-category spread is ~$5. H0 rejected statistically. Business implication: Category-based differential pricing would not produce meaningful revenue differentiation given the current uniform pricing structure.

8.7 PAYMENT METHOD VS. AMOUNT — ONE-WAY ANOVA

All four payment methods yield approximately $255 average transaction amount. F-statistic significant due to sample size. Business implication: Payment method cashback incentives may influence method preference but will not shift transaction basket size.

8.8 SHIPPING METHOD X ORDER STATUS — CHI-SQUARE TEST

Cramer's V < 0.01 despite chi-square significance. Delivered rates by shipping method: Same-Day 44.6%, Express 43.5%, Standard 42.3% — a 2.3% spread. Business implication: Operational improvements to delivery rate should not focus on shipping method selection.

8.9 MONTHLY REVENUE SEASONALITY — ONE-WAY ANOVA

F-statistic significant but peak-to-trough revenue swing across 12 months is only ~6%. Business implication: No seasonal promotional campaigns are required — inventory and staffing can be planned at constant levels year-round.

8.10 COUNTRY VS. TOTAL_AMOUNT — KRUSKAL-WALLIS TEST

All five countries show nearly identical per-transaction medians despite the USA's 31.6% revenue share. USA revenue dominance comes from transaction volume, not higher per-order spending. Business implication: Revenue growth in non-USA markets should focus on customer acquisition and transaction frequency.

SUMMARY OF STATISTICAL TESTS

Test	Method	H0 Rejected	Practical Effect
Amount x Total_Amount (Pearson r)	Pearson corr.	Yes	Large (deterministic, r=0.67)
Total_Purchases x Total_Amount (Pearson r)	Pearson corr.	Yes	Large (deterministic, r=0.65)
Age vs Ratings (Spearman rho)	Spearman corr.	Yes	Negligible (rho=0.17)
Gender vs Total_Amount	Welch's t-test	Yes	Negligible (d<0.01)
Income Level vs Total_Amount	Kruskal-Wallis	Yes	Negligible ($11 mean spread)
Customer Segment vs Total_Amount	Kruskal-Wallis	Yes	Negligible ($4 median spread)
Product Category vs Total_Amount	Kruskal-Wallis	Yes	Negligible ($5 mean spread)
Payment Method vs Amount	One-way ANOVA	Yes	Negligible (~$255 all methods)
Shipping Method x Order Status	Chi-square	Yes	Negligible (Cramer's V<0.01)
Monthly Revenue Seasonality	One-way ANOVA	Yes	Negligible (6% swing)
Country vs Total_Amount	Kruskal-Wallis	Yes	Negligible (uniform per-txn)

Key takeaway: At n = 293,468, statistical significance is achieved for almost every test due to the law of large numbers. The consistent finding across all tests is that while statistical significance is present, practical effect sizes are negligible (Cohen's d < 0.01, Cramer's V < 0.01, mean spreads of $4–$11). This strongly suggests a uniformly structured synthetic dataset where pricing, segmentation, and operations are not materially differentiated.

9. TABLEAU DASHBOARD DESIGN

Three interactive Tableau dashboards were developed using the tableau_ready_dataset.csv (293,468 rows, 33 columns). All dashboards are published on Tableau Public and linked to tableau/dashboard_links.md in the GitHub repository.

DASHBOARD 1 — SALES & REVENUE PERFORMANCE (D1)

Objective

This executive-level dashboard answers: How much revenue are we generating, where is it coming from, and is it trending in the right direction? It is designed for the CCO or CEO to monitor top-line performance across time, geography, and category.

Views & Charts

- KPI Banner (top): Total Revenue ($401.6M), Revenue per Customer ($4,641.45), Total Orders (293,468), Avg Units per Order (5.36), Avg Order Value ($1,369). Global filters: Product Category, Country, Payment Method.
- Revenue Trend Line (combo chart): Monthly bar chart for Total Revenue with a transaction count line overlay. Reveals the stable, non-seasonal revenue pattern across all 12 months.
- Map by Country (treemap): USA ($126.92M), UK ($84.59M), Germany ($70.20M), Australia ($60.13M), Canada ($59.80M). Size and shade encode revenue magnitude.
- Revenue by Category (horizontal bar): Electronics ($95.19M), Grocery ($88.78M), Clothing ($72.86M), Books ($72.55M), Home Decor ($72.23M). Near-equal bar lengths visualise the uniform revenue distribution.
- Revenue Share by Country (pie): USA 31.6%, UK 21.1%, Germany 17.5%, Australia 15.0%, Canada 14.9%.
- AOV by Payment Method (bar): Credit Card ($119.84M total), Debit Card ($102.87M), Cash ($97.10M), PayPal ($81.80M) — reflecting transaction count differences rather than AOV differences.

DASHBOARD 2 — CUSTOMER BEHAVIORS & SEGMENTATION (D2)

Objective

This dashboard answers: Who are our customers and how do their characteristics (age, income, gender, segment) influence spending? It supports the customer's success team and marketing leadership.

Views & Charts

- KPI Banner: Avg Spend per Customer ($4,641.45), Unique Customers (86,529), Avg Purchase Frequency (3), Premium Customer Count (44,650), High Income Revenue % (24.8%), New vs Returning Split (65%). Filters: Customer Segment, Income, Age Group.
- Revenue by Segment (horizontal bar): Regular ($195.9M), New ($121.2M), Premium ($84.6M). Highlights the Regular segment as the revenue backbone.
- Revenue by Income (donut): Low, Medium, High segments with near-equal shares — confirming the statistical finding of income-neutral spending.
- Spend by Age Group (bar): 18–23 age group shows the highest total spend due to transaction volume; all groups show similar per-transaction values.
- Gender Revenue (grouped bar by segment): Male customers dominate revenue within every segment, consistent with the 62.2% male transaction share.
- Revenue by Segment & Income Level (grouped bar): No meaningful interaction effect — all segment-income combinations show similar revenue, confirming statistical independence.
- Customer Count by Country + Segment (stacked bar): USA leads all countries in customer count across all three segments; composition is similar across countries.

DASHBOARD 3 — OPERATIONS & CUSTOMER SATISFACTION (D3)

Objective

This operational dashboard answers: How satisfied are our customers, how are orders being fulfilled, and is there any relationship between shipping method, product category, and satisfaction? Targets the operations and customer experience teams.

Views & Charts

- KPI Banner: Positive Feedback % (64.98%), Avg Rating (3.164), Delivered Order Rate (43.52%), Same-Day Delivery Rate (44.62%), Pending Order Rate (16.14%). Filters: Order Status, Shipping Method, Payment Method, Country, Product Category.
- Avg Rating by Shipping Method (bar): Same-Day (3.20) > Standard (3.14) > Express (3.15). The difference is minimal (~0.06), confirming shipping method has a negligible satisfaction impact.
- Category x Shipping (heat matrix): Electronics with Same-Day shipping shows the highest rating (3.39). All other cells range narrowly from 3.10–3.22. Electronics + Same-Day is the highest-satisfaction combination.
- Rating Trend Over Time (line): Ratings are flat across all 12 months (3.15–3.18), with a slight uptick in October (3.18). No actionable seasonality in satisfaction.
- Feedback Category Distribution (bar): Excellent (~98K), Good (~87K), Average (~60K), Bad (~42K). 64.98% of transactions receive positive feedback (Excellent + Good).
- Order Status Breakdown (stacked bar by shipping method): Pending and Processing rates are consistent across Express, Same-Day, and Standard (~13–15% pending each) — confirming shipping method does not materially affect fulfilment outcomes.

10. INSIGHTS SUMMARY

The following 10 insights are stated in decision language — not as data descriptions, but as business conclusions that inform action.

INSIGHT 1 — THE USA LEADS ON VOLUME, NOT VALUE

The USA accounts for 31.6% of total revenue ($126.92M) but has the same per-transaction median ($1,042) as Canada or Australia. The revenue gap is purely driven by 93,082 USA transactions vs ~44,000 in each other market. Growing non-USA transaction counts — not premiumising USA pricing — is the lever for global revenue balance.

INSIGHT 2 — ELECTRONICS IS THE HIGHEST-REVENUE CATEGORY BUT THE GAP IS RAZOR-THIN

Electronics leads with $95.19M vs. Home Decor's $72.23M — a $23M gap. However, the average transaction value difference between Electronics and the lowest category (Grocery) is only $4.58. Electronics' revenue lead comes from transaction count (69,443 vs 52,782), not from premium pricing. Category-specific pricing strategies would have negligible impact.

INSIGHT 3 — PREMIUM CUSTOMERS ARE THE MOST SATISFIED BUT NOT THE BIGGEST SPENDERS

Premium segment customers have a 56.2% satisfaction rate — 7 percentage points higher than New customers (50.5%) and 11 points higher than Regular (45.2%). Yet Premium customers have the lowest median transaction value ($1,040). This indicates that the Premium segment is built on loyalty and experience quality, not transaction size — making retention and satisfaction investment in this segment high-ROI.

INSIGHT 4 — RETAIL SHOWS NO SEASONAL PATTERN, ENABLING CONSTANT-LEVEL PLANNING

Monthly revenue oscillates within a $2.2M band ($32.1M–$34.3M) throughout the entire year. One-way ANOVA confirms no statistically meaningful monthly variation beyond noise. This eliminates the need for seasonal staffing surges, promotional peaks, or inventory hoarding — enabling leaner, flatter operational planning.

INSIGHT 5 — PEPSI DOMINATES BRAND REVENUE THROUGH VOLUME, NOT PRICING

Pepsi generates $40.7M in total brand revenue — nearly double Samsung ($24.7M) — driven by high-frequency Grocery transactions. Pepsi's per-unit Amount is not higher than other brands; it wins on basket depth (Total_Purchases). This suggests that consumables category volume is the most accessible lever for brand-level revenue growth.

INSIGHT 6 — SHIPPING METHOD HAS NO MEANINGFUL IMPACT ON SATISFACTION OR DELIVERY RATE

Same-Day, Express, and Standard shipping methods all deliver ~43-45% of orders and generate ratings within 0.06 of each other (3.14–3.20). Cramér's V < 0.01 confirms negligible association between shipping choice and order outcomes. Premium shipping upgrades are unlikely to improve customer satisfaction or fulfilment rate at a material level.

INSIGHT 7 — PAYMENT METHOD DOES NOT INFLUENCE BASKET SIZE

All four payment methods (Cash, Credit Card, Debit Card, PayPal) produce an average transaction amount of approximately $255 per item. ANOVA confirms no practical difference. Payment method of cashback programmes or surcharge structures would not shift customer spending levels.

INSIGHT 8 — 16.14% OF ORDERS ARE PENDING, REPRESENTING A MEANINGFUL PIPELINE RISK

With 293,468 total orders, a 16.14% pending rate equals ~47,394 orders in an unresolved state. If even 25% of pending orders eventually cancel, that represents approximately $12.2M in at-risk revenue. Operational focus on converting pending to processed or shipped should be a priority.

INSIGHT 9 — THE ELECTRONICS + SAME-DAY COMBINATION DELIVERS THE HIGHEST CUSTOMER SATISFACTION

The Category x Shipping heat matrix reveals Electronics with Same-Day shipping produces an average rating of 3.39 — the only cell in the matrix that materially outperforms the dataset average (3.164). Promoting Same-Day shipping for Electronics purchases could lift the overall satisfaction rate.

INSIGHT 10 — CREDIT CARD DOMINATES PAYMENT VOLUME, SUGGESTING DIGITAL-FIRST CUSTOMERS

Credit Card accounts for 29.8% of all transactions (87,444), with Debit Card (25.6%) and Cash (24.2%) not far behind. The near-even split across all payment methods suggests a customer base that spans both digital-native and cash-preferred profiles, particularly relevant for expansion into markets with lower card penetration.

11. RECOMMENDATIONS

RECOMMENDATION 1 — ACCELERATE CUSTOMER ACQUISITION IN AUSTRALIA AND CANADA

Insight link: Insights 1 and 10.

Australia and Canada each generate ~$60M in revenue from ~44,000 transactions — nearly identical per-transaction values to the USA ($1,366–$1,372 vs USA $1,363). However, transaction volumes are less than half the USA's 93,082. These markets are not lower-value markets; they are simply lower-volume. A targeted digital acquisition campaign (social, search, affiliate) in major Australian and Canadian cities could add 20,000 new transactions per market annually.

Expected business impact: At the dataset average transaction value ($1,369), 20,000 additional transactions per market = ~$27.4M additional annual revenue per market, or ~$54.8M combined — a 13.6% total revenue increase with no change to pricing or product mix.

Feasibility: Moderate. Requires marketing investment and local payment/logistics partnerships, but existing product catalogue and pricing need not change.

RECOMMENDATION 2 — LAUNCH A PREMIUM SEGMENT FAST-TRACK PROGRAMME

Insight link: Insights 3 and 5.

Premium customers are the most satisfied segment (56.2% satisfaction rate) but represent only 21.1% of transactions and 21.1% of revenue. The Regular segment (48.7% of transactions) is the largest by volume but has the lowest satisfaction rate (45.2%). If 10% of Regular customers could be converted to Premium tier through a structured loyalty milestone programme (e.g., 10 purchases = tier upgrade + early access to new products), satisfaction rates would rise and churn risk would fall.

Expected business impact: Converting 14,295 Regular customers (10%) to Premium at the same transaction frequency would shift ~$19.6M in annual revenue into the highest-satisfaction cohort, reducing churn risk on that revenue pool by an estimated 15–20%.

Feasibility: High. This is a CRM/loyalty programme change requiring minimal product investment. Tier thresholds, rewards, and communication can be implemented within existing infrastructure.

RECOMMENDATION 3 — TARGET SAME-DAY DELIVERY BUNDLING WITH ELECTRONICS

Insight link: Insight 9.

Electronics + Same-Day shipping is the only product-shipping combination that meaningfully outperforms the dataset average rating (3.39 vs 3.164). A 'Same-Day Guarantee' bundle for Electronics purchases above $500 could be offered at a small premium ($5–$15), improving both satisfaction and per-order revenue.

Expected business impact: Electronics generates 69,443 transactions annually. If 30% of Electronics transactions above $500 were upsold to a Same-Day bundle at $10 premium, this would add approximately $208K in ancillary revenue while lifting the overall customer rating average.

Feasibility: High. Same-Day infrastructure already exists (44.62% of all deliveries are Same-Day). A bundle SKU and checkout nudge are the only required additions.

RECOMMENDATION 4 — OPERATIONALLY ADDRESS THE 16% PENDING ORDER RATE

Insight link: Insight 8.

The 16.14% pending order rate (~47,394 orders) represents significant at-risk revenue. Assuming an average total amount of $1,369, the pending pipeline holds approximately $64.9M in unconfirmed revenue. An automated order resolution workflow — triggered by 48-hour pending status — should escalate to customer success teams for proactive follow-up.

Expected business impact: Converting 30% of pending orders to complete (an achievable target for a proactive escalation programme) would protect approximately $19.5M in revenue and reduce the pending order backlog by ~14,218 orders.

Feasibility: High. This is an operational process change with minimal technology investment. Existing CRM tools can be configured to trigger escalation of workflows.

12. IMPACT ESTIMATION

Recommendation	Lever	Est. Revenue Impact	Timeline
R1 — Australia & Canada Acquisition	Volume	+$54.8M/year	12–18 months
R2 — Premium Fast-Track Programme	Retention / Segment Mix	+$3–5M/year (churn reduction)	6–12 months
R3 — Electronics + Same-Day Bundle	AOV uplift	+$208K/year	1–3 months
R4 — Pending Order Resolution	Revenue recovery	+$19.5M (one-time recovery)	1–2 months

Collectively, these four recommendations could generate $74–77M in additional or recovered revenue over an 18-month horizon — representing approximately 18.5% uplift on the $401.6M baseline — with moderate implementation cost and high strategic alignment.

13. LIMITATIONS

- Synthetic data: The dataset is generated, not from a live retail system. Statistical patterns — particularly the near-zero practical effect sizes across all tests — likely reflect the uniform data generation logic rather than real-world retail dynamics.
- Partial 2024 data: The 2024 dataset covers only January and February. Year-on-year trend analysis is not valid without normalisation, and seasonal patterns for Q2–Q4 2024 cannot be assessed.
- Ratings and Feedback redundancy: The two variables are near-deterministically linked. Models or analyses using both would suffer from multicollinearity. Only one should be used in predictive modelling.
- No cost data: The dataset contains only revenue-side metrics (Total_Amount). Without COGS, fulfilment costs, or customer acquisition costs, profitability analysis is not possible.
- No customer journey data: Transactions are analysed independently. Repeat purchase rates, time between purchases, and churn events are not observable in the current structure.
- No real-time capability: The Tableau dashboard is a static snapshot. Any operational recommendation requires integration with live data pipelines for ongoing monitoring.
- Geographic aggregation: Country-level analysis masks city-level variation. The top 10 cities account for 31.3% of revenue, but city-level segmentation is not available in the current dashboard design.

14. FUTURE SCOPE

- Predictive revenue model: With cost and churn data, a regression or gradient boosting model could be trained to predict Total_Amount or customer lifetime value (CLV) at the transaction level.
- Customer churn model: Incorporating time-between-purchases and last-purchase-date features would enable a churn propensity score for each customer, supporting proactive retention outreach.

- Real-time dashboard: Connecting Tableau (or an equivalent BI tool) to a live database with streaming transaction data would convert this static analysis into an operational monitoring system.
- NLP on product names: The 'products' column contains free-text product names (e.g., 'Cycling shorts', 'Plasma TV'). Named entity recognition (NER) or topic modelling could extract sub-category clusters for finer-grained product analysis.
- Supply chain integration: Adding inventory, reorder, and logistics cost data would enable margin analysis and fulfilment efficiency modelling — elevating the project from revenue analytics to profitability analytics.
- A/B testing framework: The current data cannot support causal inference. A controlled experiment design (e.g., testing Same-Day bundle pricing) would allow evidence-based validation of Recommendations 2 and 3.

15. CONCLUSION

This project applied a complete data analytics pipeline — from raw data extraction through Python-based ETL, exploratory and statistical analysis, and Tableau visualisation — to a 302,010-row global retail dataset. After rigorous cleaning, 293,468 validated transactions across 5 countries, 5 product categories, and 3 customer segments were analysed using 11 statistical tests and visualised across 3 interactive Tableau dashboards.

The central finding of this analysis is paradoxical but important: a $401.6M retail operation shows uniform pricing, negligible category differentiation, and no seasonal patterns across virtually every dimension tested. While this is partly an artefact of the synthetic data generation, it carries a real business message — revenue growth in such an environment must come from volume expansion and customer experience investment, not pricing strategy or product-mix optimisation.

The highest-impact recommendations are: (1) acquire more customers in Australia and Canada, where per-transaction value already matches the USA; (2) accelerate the path from New to Premium segment through structured loyalty milestones; and (3) operationally resolve the $64.9M pending order pipeline through automated escalation. Together, these actions represent a realistic $74–77M revenue opportunity over an 18-month horizon.

The Tableau dashboards built in this project provide retail leadership with a reusable, filterable view of performance across time, geography, category, and customer segment — enabling the transition from reactive reporting to proactive, data-driven decision-making.

APPENDIX A — DATA DICTIONARY

The following table defines all 33 columns in the final tableau_ready_dataset.csv.

Column	Data Type	Description
Transaction_ID	Int64	Unique transaction identifier
Customer_ID	Int64	Anonymised unique customer identifier
City	Category	City of the transaction
State	Category	State/Province of the transaction
Country	Category	Country of the transaction (USA, UK, Germany, Australia, Canada)
Age	Int64	Customer age (18–70)
Gender	Category	Customer gender (Male / Female)

Income	Category	Income bracket: Low, Medium, High
Customer_Segment	Category	Loyalty tier: New, Regular, Premium
Date	datetime64	Transaction date (YYYY-MM-DD)
Year	Int64	Year of transaction (2023 or 2024)
Month	Object	Month name (January–December)
Time	Object	Time of transaction (HH:MM:SS)
Total_Purchases	Int64	Number of items in the transaction (1–10)
Amount	Float64	Per-item price ($10–$500)
Total_Amount	Float64	Total transaction value (Amount x Total_Purchases)
Product_Category	Category	High-level category: Electronics, Grocery, Clothing, Books, Home Decor
Product_Brand	Category	Brand name (e.g., Nike, Samsung, Pepsi)
Product_Type	Category	Sub-category product type (e.g., Tablet, Shorts, Fiction)
Feedback	Category	Qualitative feedback: Excellent, Good, Average, Bad
Shipping_Method	Category	Shipping choice: Express, Same-Day, Standard
Payment_Method	Category	Payment type: Cash, Credit Card, Debit Card, PayPal
Order_Status	Category	Fulfilment status: Delivered, Shipped, Processing, Pending
Ratings	Int64	Numeric rating 1–5
products	Category	Specific product name (free text)
Month_Num	Int32	Numeric month (1–12) for Tableau sort order
Quarter	Object	Fiscal quarter: Q1, Q2, Q3, Q4
Day_of_Week	Object	Day name: Monday–Sunday
Hour	Int32	Hour of transaction (0–23)
Revenue_per_Purchase	Float64	Total_Amount / Total_Purchases
High_Value_Flag	Int64	1 if Total_Amount > $2,031.33 (75th percentile), else 0
Satisfied_Flag	Int64	1 if Ratings >= 4, else 0
Feedback_Score	Float64	Numeric encoding: Excellent=3, Good=2, Average=1, Bad=0

APPENDIX B — ETL CLEANING LOG SUMMARY

Cleaning Step	Records Affected	Action Taken

Columns dropped (PII)	5 columns	Name, Email, Phone, Address, Zipcode removed
Full duplicate rows	4 rows	Dropped (keep first)
Duplicate Transaction_IDs	7,544 rows	Dropped (keep first per Transaction_ID)
Rows with missing Transaction_ID	~1 row	Dropped
Rows with missing Customer_ID	~296 rows	Dropped
Rows with missing Amount	~347 rows	Dropped
Age imputation	169 rows	Filled with median (32)
Ratings imputation	180 rows	Filled with median (3)
Categorical mode imputation	~282–330 rows/col	Mode fill for 12 categorical columns
Total_Amount algebraic fill	695 partial rows	Derived from Amount x Total_Purchases
Rows with missing Date	351 rows	Dropped
Time null fill	336 rows	Filled with '00:00:00'
Outliers (Total_Amount > IQR)	3,777 rows	Retained (assessed as legitimate)
Final dataset shape	293,468 rows x 33 cols	Validated: 0 nulls, 0 duplicates

APPENDIX C — CONTRIBUTION MATRIX

The table below documents each team member's contribution across all project phases. Claims are verifiable through GitHub Insights, PR history, and committed notebook files.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Ananya Narang - 2401020087	✓		✓		✓		
Utkarsh Jain - 2401020071	✓			✓	✓		
Anusha Prathapani - 2401010344	✓					✓	
Rishik Chowdary Karuturi - 2401010381	✓	✓		✓			
Atanu Adhikari – 2401010111	✓		✓				✓
Shubham Aggarwal - 2401010451	✓	✓					✓

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead Signature: Ananya Narang Date: 28/04/2026